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MILITARY SPECIFICATION

GUN, TWIN 40MM, SELF-PROPELLED M42 AND M42A1
(OVERHAUL)

1. SCOPE

1.1 Scope. This specification covers overhaul of two types full-tracked, armored combat vehicles, which mounts twin 40MM guns as primary weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Federal
VV-G-76 - Gasoline, Automotive.

Military
MIL-P-514 - Plates: Identification, Instruction and Marking, Blank.
MIL-G-3056 - Gasoline, Automotive, Combat.
MIL-H-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Testing.
MIL-S-12158 - Sealing Compound Non-Curing, Polybutene.
MIL-G-45178 - Gun, Self-Propelled, Full Tracked: Twin 40MM, M42 and M42A1; Preparation for Storage and Shipment of.

STANDARDS
Federal
FEDERAL STD. NO. 595 - Colors.

Military
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-193 - Painting Procedures Tactical Vehicles (Tracked and Wheeled).

FSC-2350

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MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642 - Identification Marking of Combat, and Tactical Transport Vehicles.

PUBLICATIONS

Engineering Parts Lists

7968000 - Gun, Twin 40MM, Self-Propelled M42.
8750204 - Gun, Twin 40MM, Self-Propelled M42A1.

Master Lists

7968000 - Master List of Supplementary Quality Assurance Provisions for Gun, Twin 40MM, Self-Propelled M42.
8750204 - Master List of Supplementary Quality Assurance Provisions for Gun, Twin 40MM, Self-Propelled M42A1.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First overhaul vehicle. The first overhaul vehicle which shall be fully representative of vehicles proposed to be furnished under the contract, shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.1.1 Workmanship and reliability. The workmanship shall be of a quality to assure that the vehicle and components thereof are free from any defects that compromise, limit, or reduce the vehicle or component capabilities herein expressed. Vehicles overhauled in accordance with this design shall be subjected to examinations and tests to prove vehicle capabilities to perform in accordance with the following:

(a) Vehicle shall require only organizational maintenance (first and second echelon) during 1,600 miles of normal operation.
(b) Vehicle shall require no replacement or major overhaul of any major automotive component during 3,200 miles of normal operation.
(c) Replacement of track shoe assemblies shall not exceed 10 percent of the full track during 2,000 miles of normal operation.

3.1.1.1 Vehicle operation. The 3,200 miles of vehicle operation combat loaded or loaded with simulated load of equal weight shall consist of the following:

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(a) 25 percent on paved roads, either concrete or asphalt or any combination of the two.
(b) 25 percent on gravel and dirt roads with at least 10 percent of this distance under mud conditions.
(c) 25 percent on level cross-country.
(d) 25 percent on hilly cross-country.

3.2 Materials.

3.2.1 Ozone resistance. All new rubber components which are under tension or which may be flexed shall be ozone resistant to the degree required by the "C_1", test requirement of MIL-STD-417.

3.2.2 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.3, and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed (by part or drawing number as QPL items) on the QPL in the referenced specification or which have been approved for inclusion on such lists. Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.3 Overhaul. Vehicles, components, sub-assemblies, and assemblies shall be overhauled and assembled into a complete vehicle in accordance with contract, Engineering Parts Lists (EPL) 7968000, 8750204 and overhaul technical data package. All parts requiring identification shall be identified in accordance with MIL-STD-130.

3.3.1 Sealing.

3.3.1.1 Sealer. The sealer conforming to MIL-S-12158, shall restrict the entrance of water into the hull when vehicle is fording streams or rivers during cross country operation and when operating in deep water.

3.3.1.2 Seals. Seals shall restrict the entrance, into the bearings of all foreign matter when fording shallow or deep water. All seals shall restrict the leaking of lubricants from the bearings (see 6.4 b).

3.3.1.3 Shroud seal. The space between the upper metal surface of the engine shroud and the sealing surface of the top deck grille shall restrict the recirculation of warm air.

3.3.1.4 Hatch seals. All hatch seals shall be free from damage, deterioration, and when installed shall restrict any leakage of water into the crew compartment (see 6.4 b).

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3.3.2 Stowed equipment. The spaces provided for the On Vehicle Equipment (OVE) items shall accept and hold in place during all conditions of vehicle operation.

3.3.3 Engine. The engine shall not have excessive vibration at idle or higher speeds with engine under load. During operation of the vehicle involving all gear and speed ranges, the engine shall indicate satisfactory function. The engine shall be free of manifold, fuel and lubricant leaks.

3.3.3.1 Cooling system. With vehicle operating in ambient temperature of 115°F, the cooling system shall be capable of maintaining engine and transmission oil temperatures between 180 and 225°F, and between 200 and 285°F respectively. Oil temperatures shall be measured at the point of entry to the heat exchanger.

3.3.4 Fuel system. During engine operation, the fuel system shall maintain fuel supply to the engine when ascending and descending 60 percent grades in forward and reverse gear, and when vehicle is being operated on 30 percent side slopes, with each side of vehicle up slope.

3.3.4.1 Fuel tanks. Provisions shall be made to assure the internal cleanliness of the fuel tanks prior to initial fueling of vehicle.

3.3.4.2 Fuel and hydraulic lines. Provisions shall be made to assure internal cleanliness of fuel and hydraulic lines, and connections and tanks shall be free from leakage.

3.3.4.3 Heater. The heater shall be capable of starting and operating satisfactorily at a temperature of minus 65°F., when burning gasoline conforming to MIL-G-3056. The supply lines from the vehicle fuel tanks, separate fuel pump, and filter shall furnish fuel to the heater.

3.3.4.4 Throttle linkage. With throttle linkage attached to engine, throttle pedal fully depressed, the throttle shall be at full throttle position. With the throttle at full throttle position, the throttle pedal shall be in contact with pedal stop.

3.3.5 Exhaust system. The exhaust system shall prevent the concentration of carbon monoxide in the occupied portions of the vehicle, in excess of 0.01 percent under any condition of vehicle operation.

3.3.6 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation under all vehicle operating conditions.

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3.3.7 Adjustment mechanisms. All adjustment mechanisms shall function properly, and maintain adjustment during all vehicle operating conditions.

3.3.7.1 Driver's and commanders hatch. The driver's and commanders hatch shall break away and slide through its full range of travel without undue effort or indications of binding or interference. Hatch shall lock securely in the open and closed position throughout all vehicle operating conditions.

3.3.7.2 Front door. Front door shall be properly aligned and operate through its full range of travel without undue effort or indications of binding, and shall lock securely in the open and closed position. Seal shall be free of damage and deterioration, and when installed shall restrict entrance of water into vehicle (see 6.4 b).

3.3.7.3 Driver's seat. All sliding parts and operating contact surfaces shall be free of paint. The driver's seat shall operate through its horizontal and vertical adjustments without undue effort or indications of binding. The driver's seat shall move forward, backward, vertically or dump when the appropriate levers are activated.

3.3.7.4 Front and rear drain valves. Drain valves and controls shall operate freely, drain valves shall open and close without indications of binding or sticking. Drain valve controls must be capable of locking drain valves in the open position.

3.3.7.5 Air cleaner. All components of the air cleaner system as installed in the vehicle shall be in good condition, and correctly installed.

3.3.8 Vision devices and receptacles. Receptacles incorporated in the hull, including driver's hatch, shall accept applicable vision devices (normally installed in the field), without binding or interference. Locking devices shall function properly and secure vision device installed in the receptacle. When installed, the vision devices in the receptacle shall restrict entrance of water into the vehicle (see 6.4 b).

3.3.9 Fire extinguisher system. State of charge of each cylinder shall be properly recorded. Release mechanism shall actuate the system without undue effort on the release handle. System shall discharge properly and shall be free of CO_2 leaks. All items and components of the fire extinguisher system shall be correctly installed.

3.3.10 Ammunition racks. The hull ammunition stowage racks shall be free of all defects and conditions adversely affecting proper stowage and retention of ammunition.

3.3.11 Turret.

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3.3.11.1 Gunner's seats. All sliding parts and operating contact surfaces shall be free of paint. The seat shall be free of binding and interference throughout its operating range. Locking devices shall function properly and shall hold seat in the desired position throughout all operating conditions. CAUTION: Operator shall be sitting in seat when it is raised or lowered.

3.3.11.2 Loaders seat. With the seat properly installed and not in use, the lifting spring shall be capable of moving and retaining the seat in a vertical position.

3.3.11.3 Turret ammunition racks and boxes. The turret ammunition racks and boxes shall be free of all defects and conditions adversely affecting proper stowage and retention of ammunition.

3.3.11.4 Cartridge case chutes and deflectors. The chute and deflectors shall be free of all defects and conditions adversely affecting proper ejection of cartridge case.

3.3.11.5 Turret components. Turret components shall be correctly installed.

3.3.11.6 Foot control firing mechanism. The foot control firing mechanism shall be correctly installed. When depressing the foot pedal there shall be no binding or excessive free travel in linkage.

3.3.11.7 Elevating and traversing locks. The elevating and traversing locks shall engage and disengage without interference and shall lock the gun and mount securely under all vehicle operations.

3.3.12 Welding repairs. Welding repairs of armor or attachment to armor shall be made only when, and to the extent, specifically authorized by the procuring activity.

3.3.13 Electrical system. Each circuit shall maintain continuity from end to end without any shorts from pin to other pins and from each pin to the connector shell.

3.3.13.1 Generator voltage. The engine shall be operated at 1000 revolutions per minute (rpm) for 30 minutes before setting voltage regulator. With the vehicle generator operating at a nominal temperature, it shall deliver between 27.8 and 28.3 volts "dc" at regulator under all conditons of vehicle operation.

3.3.13.2 Slip ring (hull to turret) and external communication circuits. The slip ring and external communication circuits shall maintain continuity without evidence of shorts. Each circuit shall maintain continuity from end to end, without any shorts from one pin to the other and from each pin to the connector shell.

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3.3.13.3 Main gun firing circuits. The main gun firing solenoid shall provide the energy necessary to detonate main round.

3.3.13.4 Lights. The internal and external lights as specified on applicable drawings shall operate as specified through all vehicle operating conditions. All electrical contacts and connections shall maintain positive contact under all vehicle operating conditions.

3.3.13.5 Auxiliary generator. The auxiliary generator consisting of an engine, generator, and controls shall operate as starter generator combination. With generator operating in nominal ambient temperature, it shall deliver between 27.8 or 28.3 volts "dc" at regulator under all conditions of vehicle operation.

3.3.13.6 Personnel heaters and bilge pump. Personnel heaters and bilge pump shall function properly in starting, operating and stopping.

3.3.13.7 Infrared power pack high voltage supply. With the generator supplying specified voltage, the infrared power pack with an actual load or simulated load, output voltage at the connector shall be a minimum of 16,000 volts, without any evidence of high voltage arcing.

3.4 Vehicle break-in run.

3.4.1 Preparation. Prior to break-in run, each vehicle shall be completely assembled and properly lubricated as specified by lubrication order. Fuel tanks shall be filled to the level necessary to complete the break-in run and the required performance testing of the vehicle. Fuel used shall conform to VV-G-76 or MIL-G-3056. Engine governor shall be set to provide operation of the engine between 2800 and 2900 RPM at full load, full throttle. The no-load operation shall be limited to 3000 RPM. The vehicle shall be examined for fuel and lubricant leaks. All other necessary adjustments and safety checks shall be completed.

3.4.2 Oil pressure and temperature. Prior to the start of the break-in run, the engine shall be operated at 800 to 1500 rpm until lubricating oil is at specified operating pressure and temperature. If at any time during the break-in run the engine has been stopped for at least 30 minutes, the engine shall again be operated, as above, before continuing break-in run. Proper oil pressure and temperature shall be maintained during operation on level ground, 60 percent grades and 30 percent side slope.

3.4.3 Operation and distance. Prior to tests, each vehicle shall be given a break-in run for the distances specified in Table I on smooth, level, hard-surfaced roads or test track. The vehicle shall be operated over the undulating section of road (see figure 1) during "a" and "b" divisions of the break-in run, twice in each direction.

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Table I. Speeds and distances for break-in run

Division of run	Speed mph	Distance miles	Test conditions
a	0 to 10	5	Track (road)
b	11 to 20	7 1/2	Track (road)
c	21 to Max. Governed	7 1/2	Track (road)

3.4.3.1 Reverse operation. After each division of the break-in run, vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet.

3.4.4 Condition after run. After completion of the break-in run, the engine shall run smoothly when operated between 750 and 900 rpm's. Prior to vehicle being submitted to the Government for acceptance, all suspension wheel bearings shall be checked to determine if adjustments are correct and readjusted as required. There shall be no defects indicative of damaged components, or maladjustments that may cause faulty vehicle operation.

3.5 Performance. A complete vehicle, combat loaded (see 3.3.2) and serviced shall perform as specified herein. Ammunition may be simulated in size and weight. Vehicle, serviced (see 3.4.1) and equipped for existing climatic conditions, shall operate as specified without special equipment. Vehicle shall be capable of operating on 60 percent longitudinal slopes, headed up and headed down slopes, and on right and left 30 percent side slopes, with each side of vehicle up slope.

3.5.1 Environmental. The vehicle shall operate in ambient temperature of plus 115°F to minus 25°F, and with special equipment installed at temperature range of minus 25°F to minus 65°F.

3.5.2 Power package and power train. When power package is installed in accordance with overhaul technical data package, it shall function throughout all gear and speed ranges without loss of lubricants, grabbing, and chattering or slippage when controls are applied to control, stop, and hold vehicle.-

3.5.3 Speeds.

3.5.3.1 Level road speeds. The vehicle shall operate at sustained speeds of 25 mph and shall be capable of attaining not less than 40 mph, and at 2.5 mph on smooth, level relatively hard-surfaced roads without damage to power plant and power train. When traveling between 15 and 30 mph the vehicle drift shall not exceed 3 feet in 100 feet.

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3.5.3.2 Grade speeds. The vehicle shall operate at a sustained speed of 10 mph while ascending a 10 percent grade, and at 20 mph while ascending a 3 percent grade.

3.5.4 Acceleration. The vehicle shall accelerate from a standing start on level ground to 20 mph in not more than 12 seconds.

3.5.5 Engine starting on grades and slopes. When standing on a 60 percent grade, for not less than 2 minutes, with engine operating under no load between 700 and 750 rpm, the engine shall be stopped for not less than 2 minutes. The engine shall restart in not more than 1 minute when headed up and headed down grade. Similar engine stopping and starting shall be accomplished on 30 percent right and left side slopes.

3.5.6 Braking and drift.

3.5.6.1 Stopping. Vehicle traveling at 20 mph shall stop within 45 feet from point of brake application; drift not to exceed 4 feet when stopping.

3.5.6.2 Holding. With vehicle combat loaded, (see 3.3.2) or a simulated load of equal weight and standing on a 60 percent grade with service brakes applied, the vehicle shall be held stationary when headed up and headed down grade. With parking brake engaged and all other holding devices inoperative, the vehicle shall be held stationary when headed up and headed down grade.

3.5.7 Turning. The vehicle shall pivot 360 degrees to the right and left in neutral steer within a circle 35 feet in diameter.

3.5.8 Fording.

3.5.8.1 Shallow water. Vehicle shall ford a level, hard-bottom body of clean water 40 inches in depth without special equipment. With vehicle standing in water 40 inches in depth for 30 minutes, the accumulation of water shall be not more than 1 inch on the crew compartment hull floor measured forward of the bulkhead.

3.5.8.1.1 Engine operation. With the vehicle engine compartment flooded with 40 inches of clear water (from ground level) and engine operating at 1000 rpm for 15 minutes then stopped and restarted within 3 minutes. Engine and all accessories shall function satisfactorily during and after fording operations.

3.5.8.2 Lubrication contamination. All accessories shall function satisfactorily during and after fording operations. After fording operations, the water contamination content of the transmission, engine, final drives, and suspension system lubricants shall be not more than 2 percent by volume.

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3.5.8.3 Bilge pumps. The bilge pumps, installed in accordance with applicable drawings, shall remove water from the vehicle at a minimum rate of 50 gpm each, with a discharge head of 4 1/2 feet and with pump motor operating on 27.5 volts impressed at the motor.

3.5.9 Trench crossing. The vehicle shall cross trenches 36 inches or more in depth and 72 inches in width without stalling or damage to suspension, and gun tube, and fenders.

3.5.10 Vertical obstacles. The vehicle shall cross over vertical obstacles 28 inches in height without stalling or damage to suspension and hull floor.

3.5.11 Climbing. Vehicle from a standing on grade position shall ascend longitudinal grades of 60 percent in forward and reverse gear without stalling or damage to power plant and power train. Vehicle shall operate on right and left side slopes of 30 percent.

3.5.12 Vehicle turret control, gun control, and gun fire control system performance.

3.5.12.1 Turret and gun control system.

3.5.12.1.1 Manual traverse effort. With the gun horizontal and mount or carriage level, the traversing mechanism shall operate smoothly and with uniform effort. The torque required at hand crank shaft to traverse the mount or carriage in either direction shall not exceed 90 inch pounds.

3.5.12.1.2 Traverse mechanism backlash. Backlash in the traversing mechism shall be as follows: With the gun approximately horizontal and mount or carriage level, the backlash shall not exceed 1/2 mil at the gun muzzle, corresponding to 2 degress and 5 minutes at the oil gear shaft.

3.5.12.1.3 Power traverse. With 27.5 volts applied at slip ring the control system M16A1E1 shall operate the turret clockwise and counterclockwise through 360 degrees traverse without excessive surge or erratic movement. With control handle in full slew position, the traverse speed shall be not less than one complete revolution of turret in 9 seconds. Rate of movement shall be the same both directions.

3.5.12.1.4 Elevation and traverse system stability. When turning rate motor toggle switch on drive controller to the ON position the rate motor should start immediately. With rate motor running and drive controller grip assemblies in neutral position, the amount of creep or gun mount movement shall exceed 675 mils per minute in azimuth or 3 degrees per minute in elevation.

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3.5.12.1.5 Manual elevation effort. The torque required at the hand crank shaft, to elevate or depress the gun, shall not exceed 90 inch-pounds between 5 degree elevation and 80 degree elevation. Torque shall not exceed 125 inch pounds at points below 5 degrees elevation and above 20 degrees elevation. Elevating mechanism shall operate smoothly through full range of elevation.

3.5.12.1.6 Elevation mechanism backlash. Backlash in the elevating mechanism shall be as follows: With the gun approximately horizontal and mount or carriage level, the backlash shall not exceed 1/2 mil at the gun muzzle, corresponding to 3 degrees and 17 minutes at the oil gear shaft.

3.5.12.1.7 Manual elevation limit stops. The elevation limit stops shall be set to provide verticle movement of the dual gun from -3° to $+87^{\circ}$.

3.5.12.1.8 Power elevation. Elevate and depress the guns by rotating the grip assemblies forward for depression or backward for elevation. Gun movement shall be smooth without erratic action. The hydraulic limit stop assembly in elevation oil gear shall limit gun movement to minus 3° and plus 85°. The guns shall elevate from a position of 20° to a position of 80° in 2.4 seconds and depress from a position of 60° to a position of 0° in 2.4 seconds.

3.5.13 Gun fire control system.

3.5.13.1 Synchronization. The sight computer M38 leveling pads shall be parallel to the gun mount carriage leveling pads within $\pm$ 2 mils from maximum depression to 45 degree elevation. From elevations of 45 degree to maximum elevation, leveling pads of sight computer M38 and leveling pads of gun mount carriage shall be parallel within $\pm$ 4 mils.

3.5.13.2 Boresighting. The boresighting lines of sight of the sighting instruments shall be in line with a line of sight through the axis of the gun bores upon a common aiming point not less than 1000 yards distant. The synchronization and boresighting of the gun and computor shall insure the accuracy of hitting target.

3.5.14 Radio suppression. Vehicle shall be radio interference suppressed in accordance with tactical vehicle requirements of CLASS III of MIL-STD-461.

3.6 Painting. Unless otherwise specified, the exterior of vehicle, components, assemblies, sub-assemblies, parts that require painting, and interior of vehicle shall be prepared and painted in accordance with MIL-STD-193. Vehicle exterior paint and exterior non-skid paint shall match color chip X24087 of Federal Std. No. 595.

3.7 Marking. Marking of vehicle shall be in accordance with MIL-STD-642.

3.8 Data plates. All data plates shall be in accordance with MIL-P-514.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Parts and components. To determine conformance to 3.3, new parts, components and assemblies (as required) shall be inspected in accordance with drawings listed on EPL 7968000 and 8750204 and the Master List of Supplementary Quality Assurance Provisions 7968000 and 8750204. The reused items shall be inspected in accordance with inspection data listed in the overhaul technical data package.

4.1.2 Contractor's quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to determine quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Qualified product. The contractor's inspection records shall be checked to determine contractor conformance to 3.2.2.

4.2 First overhaul vehicle inspection.

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4.2.1 Special process inspection. A special process inspection will be conducted by the Government representatives during overhaul of the first vehicle (see 3.1) to evaluate conformance of materials and workmanship with technical data package.

4.2.1.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection systems and plan of quality control, will be conducted on the following: Material, welding, surface hardening, heat treating, protective finishes, etc.

4.2.1.1.1 Hull inspection. The hull shall be inspected prior to covering the weldments with paint or other pertinent components or assemblies.

4.2.1.2 Complete vehicle inspection. The first vehicle when complete, shall be inspected by the Government subsequent to contractor's inspection at the contractor's plant to determine conformance to contract and specification requirements.

4.2.1.2.1 Corrections. Corrections found necessary as a result of above inspection shall be made by the contractor on the vehicle.

4.2.1.3 Overhauled production vehicle test. After completion of the first overhauled vehicle (see 3.1) inspection, a second vehicle shall be selected from the first months production or one of the first 10 shall be subjected to examination (see 4.4.1) and tests specified in 4.5 (a) and (b). Subsequent to examination and tests, the selected vehicle shall be subjected to the 3,200 mile test specified in Table II at a site approved by the Government, which will require a maximum of 90 days. These tests are to be performed by the Government. Delay in test completion caused by vehicle breakdown or failure of the contractor to adequately support the vehicle with parts during the tests shall not be the basis for adjustment of the contract, delivery schedule or contract price.

Table II. 3,200 Mile Test Combat Loaded
(Government Proving Grounds)

Course	Mileage & speeds
Hard surface roads	800 miles at varying speeds up to maximum
Gravel and dirt roads	800 miles at varying speeds up to maximum
Level cross country	800 miles at varying speeds up to maximum
Hilly cross country	800 miles at varying speeds up to maximum

4.2.2 Test failure. Any failures or deficiencies of a workmanship or materials nature found during or as a result of the 3,200 mile test, will be cause for rejection of the vehicle. Further, the Government will refuse to continue acceptance of overhauled vehicles until evidence has been provided by the contractor

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that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of 3,200 mile test shall be prima facie evidence that all vehicles already accepted prior to completion of the 3,200 mile test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 Break-in run (track). To determine conformance to 3.4.1, 3.4.2, and 3.4.3 each vehicle shall be lubricated and serviced prior to the break-in run. The run shall be on smooth, level, hard-surfaced roads, and for the distance specified in Table I. The vehicle shall be operated in reverse after each division specified in 3.4.3.1. The undulating portion of the test shall be accomplished twice in each direction at speed between 5 and 10 mph during division "a" and between 11 and 14.5 mph during division "b".

4.3.1 Condition after run. To determine conformance to 3.4.4, after completion of the break-in run, the vehicle shall be checked to determine that there is no damage or malfunction to the vehicle and that adjustments are as required.

4.4 Inspection provisions.

4.4.1 Examinations. Visual, dimensional, and primary functional examination shall be performed on each vehicle for conformance to the technical data package and this specification. Examination shall be performed against the classification of defects shown in Table III. The following constitutes a part of the classification of defects (see 6.4).

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any leak that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

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Table III. Classification of defects

Characteristic	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
Engine	101. Malfunction, leaks (see 6.4 c), mounting improper, damage	201. Improper component assembly or installation	Visual and functional
Engine governors	102. Improper setting, malfunction	202. Missing or defective seals	Functional
Power train	103. Malfunction, mounting (improper, damage, leaks (see 6.4 c)	203. Improper component assembly or installation	Visual and functional
Fuel system components	104. Malfunction, damage, leaks (see 6.4 a)	204. Improper assembly, installation or clearance	Visual and functional
Cooling system components (engine and transmission)	105. Malfunction, damage, leaks (see 6.4 b)	205. Improper assembly or installation	Visual and functional
Exhaust system components	106. Damage, leaks	206. Improper assembly, installation or clearance	Visual
Electrical system components (includes wiring and instruments)	107. Malfunction, damage	207. Improper assy., installation, circuit identification clearance or protection, illumination of reticle	Visual and functional
Suspension system components	108. Damage, tire or track pad base separation, improper clearance or alignment, welding defects, leaks (see 6.4 c), malfunction	208. Improper assy., installation, bolt or nut torque and track	Visual
Hull	109. Structure and welding defects (Metal shrink or crater cracks excluded)		Visual
Controls and adjusting mech.	110. Malfunction, improper clearance	209. Improper assembly or installation	Visual and functional
Doors, fenders, grilles, panels, ducts, stowage brackets, boxes, shocks and stops	111. Malfunction: torsion spring and hold-open devices	210. Improper fits, adjustment, assembly or installation	Visual and functional
Drain valves	112. Valve seating improper	211. Improper assy., installation of control adjustment	Visual and functional

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Table III. Classification of defects (Cont'd)

Characteristic	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
Cushions, seats, rests and crash pads	113. Seat control malfunction	212. Improper fit, installation adhesions or damage	Visual and functional
Turret traverse lock	114. Locking malfunction, damage	213. Improper assembly or installation	Visual and functional
Traversing and elevating mech.	115. Clearance improper, component damage	214. Improper assembly or installation	Visual and functional
Main armament	116. Component damage, improper assembly or trunnion cap bolt torque		Visual and functional
Weapon control system, (includes panels, solenoids switches, wiring and other components)	117. Component damage, defective wiring	215. Improper clearance, assy., installation, circuit identification, protection or markings	Visual and functional
Misc. items or accessories	118. Malfunction, damage, missing	216. Improper assembly or installation	Visual and functional
Fire extinguisher system components	119. Seals missing or defective	217. Data card incomplete, improper assembly or installation	Visual
Paint-protective and anti-skid coatings		218. Application, coverage and materials improper	Visual
Lube fittings and lubrication		219. Fittings missing, defective or improperly installed, lubrication improper	Visual
Decals, markings, data and instruction plates		220. Incomplete data, missing, improper location or site, illegible	Visual

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4.4.1.1 Unclassified defects. All component defects having no bearing on function, safety, interchangeability or life, but which are considered departures from good workmanship, will be noted in writing. Workmanship deficiencies falling within this category and recurring will be added to the minor classification of defects of this specification.

4.5 Classification of tests

(a) Acceptance tests, 4.5.1.1 through 4.5.1.32.3.
(b) Control tests, 4.5.2 through 4.5.2.7.
(c) Comparison tests (CT), 4.5.3 through 4.5.3.12.

4.5.1 Tests.

4.5.1.1 Acceptance tests. To determine conformance to 3.5, after break-in run (see 3.4), each vehicle shall be operated for a distance of 8 miles or more by the contractor and subjected to acceptance tests specified in Table IV. Except for equipment and component installation and servicing performed by the contractor as a final processing of vehicles for acceptance, vehicles shall be completely assembled, all adjustments completed, and serviced (see 3.4.1) with fuel tanks filled to capacity, lubrication and fluid as specified, but need not be combat loaded except for holding test specified in 4.5.1.28. After the tests, the vehicle shall be examined for evidence of fuel or lubricant leakage, or other deficiencies.

4.5.1.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government shall stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

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Table IV. Location for tests

Title	Paragraph No.	Place of Manufacture	Government Proving Ground
Acceptance tests	4.5.1.1	X	X
Sealer and seals test	4.5.1.2	X	X
Hatch seal test	4.5.1.3	X	X
Engine test	4.5.1.4	X	X
Fuel system slope check	4.5.1.5	X	X
Fuel and hydraulic lines check	4.5.1.6	X	X
Heater check	4.5.1.7	X	X
Throttle linkage check	4.5.1.8	X	X
Controls check	4.5.1.9	X	X
Adjustment mechanism check	4.5.1.10	X	X
Driver's and Commanders hatch and escape door check	4.5.1.11	X	X
Driver's seat check	4.5.1.12	X	X
Front and rear drain valve test	4.5.1.13	X	X
Air cleaner check	4.5.1.14	X	X
Vision devices and receptacle check	4.5.1.15	X	X
Ammunition racks check	4.5.1.16	X	X
Gunner's and loaders seat check	4.5.1.17	X	X
Turret ammunition racks, boxes, case chutes, and deflectors check	4.5.1.18	X	X
Turret components, foot control firing mechanism, elevating and traversing check	4.5.1.19	X	X
Welding repairs and burning checks	4.5.1.20	X	X
Electrical system check	4.5.1.21	X	X
Infrared power pack high voltage supply check	4.5.1.22	X	X
Power package and train test	4.5.1.23	X	X
Level road speed test	4.5.1.24	X	X
Acceleration test	4.5.1.25	X	X
Engine starting on grades and slope tests	4.5.1.26	X	X
Stopping test	4.5.1.27	X	X
Holding test	4.5.1.28	X	X
Turning test	4.5.1.29	X	X
Fording test	4.5.1.30	X	X
Turret and gun control system test	4.5.1.31	X	X
Manual traverse effort	4.5.1.31.1	X	X
Traversing mechanism backlash	4.5.1.31.2	X	X
Power traverse	4.5.1.31.3	X	X
Elevation and traverse system stability	4.5.1.31.4	X	X

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Table IV. Location for tests (Cont'd)

Title	Paragraph No.	Place of Manufacture	Government Proving Ground
Manual elevation effort	4.5.1.31.5	X	X
Elevation mechanism backlash	4.5.1.31.6	X	X
Manual elevation limit stops	4.5.1.31.7	X	X
Power elevation	4.5.1.31.8	X	X
Synchronization	4.5.1.32.1	X	X
Boresighting check	4.5.1.32.2	X	X
Boresighting procedure	4.5.1.32.3	X	X
Control tests	4.5.2	X	X
Contamination test	4.5.2.3	X	X
Stowed equipment test	4.5.2.4	X	X
Fire extinguisher test	4.5.2.5	X	X
Engine operating test (shallow water)	4.5.2.6	X	X
Bilge pump water removal check	4.5.2.7	X	X
Comparison tests	4.5.3		X
Fuel system test	4.5.3.2		X
Exhuast system test	4.5.3.3		X
Communications equipment test	4.5.3.4		X
Sub-zero temperature tests	4.5.3.5		X
High temperature tests	4.5.3.6		X
Speed on grades test	4.5.3.7		X
Trench crossing tests	4.5.3.8		X
Vertical obstacle test	4.5.3.9		X
Climbing test	4.5.3.10		X
Radio suppression test	4.5.3.11		X
Carbon dioxide distribution test	4.5.3.12		X

4.5.1.2 Sealing and seals check. To determine conformance to 3.3.1.1 and 3.3.1.2, prior to and after fording operations the vehicle shall be checked for the amount of water in vehicle, the leaking of lubricants, and 3.3.1.3 requirement checked for warm air recirculation.

4.5.1.3 Hatch seal test. To determine conformance to 3.3.1.4, the driver's loader's, and commander's hatches shall be closed and locked, and a spray of water directed on each hatch for a period of 3 minutes. All hatches shall be checked for leakage.

4.5.1.4 Engine and cooling system. To determine conformance to 3.3.3 and 3.3.3.1, the vehicle shall be operated as specified and observed for performance requirements, and the cooling system checked for maintaining specified temperatures at local ambient air temperature.

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4.5.1.5 Fuel system slope check. To determine conformance to 3.3.4, the vehicle shall be operated on grades and slopes and observed for functional requirements.

4.5.1.6 Fuel and hydraulic lines check. To determine conformance to 3.3.4.1 and 3.3.4.2, the fuel and hydraulic system shall be pressurized with dry air at a pressure of 3 to 6 psi, prior to engine installation, and checked for leaks at the fittings and fuel tank seals, by applying soapy water or equivalent.

4.5.1.7 Heater check. To determine conformance to 3.3.4.3, with the heater pump operating as specified the heater shall be checked for functional requirements.

4.5.1.8 Throttle linkage check. To determine conformance to 3.3.4.4 the throttle pedal shall be operated to full throttle, and check the engine throttle rod to insure that full throttle position has been reached.

4.5.1.9 Controls check. To determine conformance to 3.3.6, all controls shall be operated and checked for functional requirements.

4.5.1.10 Adjustment mechanism check. To determine conformance to 3.3.7, all adjustment mechanisms requiring adjustment shall be checked for proper adjustment.

4.5.1.11 Driver's and commander hatch and front door check. To determine conformance to 3.3.7.1, and 3.3.7.2, the driver's and commanders hatch and front door shall be closed, opened, then closed, and checked for binding, interference, and locking operation.

4.5.1.12 Driver's seat check. To determine conformance to 3.3.7.3, the driver's seat shall be operated through its complete range of operation and checked for functional requirements.

4.5.1.13 Front and rear drain valve test. To determine conformance to 3.3.7.4, the drain valves shall be operated, as specified, and checked for functional requirements in opened closed and locked positions.

4.5.1.14 Air cleaner check. To determine conformance to 3.3.7.5, the air cleaner shall be checked for functional and installation requirements.

4.5.1.15 Vision devices and receptacles check. To determine conformance to 3.3.8, all receptacles shall be checked for acceptability of vision devices, and for leaks.

4.5.1.16 Ammunition racks check. To determine conformance to 3.3.10, all ammunition racks shall be checked for the acceptability and retention of ammunition.

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4.5.1.17 Gunner's and loaders seat checks. To determine conformance to 3.3.11.1 and 3.3.11.2, the seats shall be operated throughout all ranges of operation and checked for functional requirements.

4.5.1.18 Turret ammunition racks, boxes, case chutes, and deflectors check. To determine conformance to 3.3.11.3 and 3.3.11.4, the racks, boxes, cartridge case chutes, and deflectors shall be checked for proper installation condition.

4.5.1.19 Turret components, foot control firing mechanism, elevating and traversing locks check. To determine conformance to 3.3.11.5 through 3.3.11.7, all items shall be checked for proper installation, conditions, damage and functional requirements.

4.5.1.20 Welding repairs and burning checks. To determine conformance to 3.3.12, all welding repairs and burning operations shall be checked for authorized approval.

4.5.1.21 Electrical system check. To determine conformance to 3.3.13, through 3.3.13.6 the electrical system shall be checked using a calibrated master voltage meter.

4.5.1.22 Infrared power pack high voltage supply check. To determine conformance to 3.3.13.7, the master battery control and infrared power switches turned on and generator supplying specified voltage, the voltage shall be measured at the loose harness plug using an 80 megohm load.

4.5.1.23 Power package and power train test. To determine conformance to 3.5 and 3.5.2, the vehicle shall be operated throughout all gear and speed ranges, the power plant, power train, final drives, tracks, and suspension system, and applicable controls checked for functional requirements.

4.5.1.24 Level road speed test. To determine conformance to 3.5.3.1, the vehicle shall be operated at maximum speed for not less than 10 minutes and at minimum speed for not less than 5 minutes and checked for function of components. Drift shall be not more than that specified.

4.5.1.25 Acceleration test. To determine conformance to 3.5.4, the vehicle shall be operated as specified, the distance and time checked.

4.5.1.26 Engine starting on grades and slopes test. To determine conformance to 3.4.2 and 3.5.5, the engine shall start and oil pressure temperature shall be maintained when operating on specified grades and slopes, The 60 percent grade test may be conducted in conjunction with brake holding test.

4.5.1.27 Stopping test. To determine conformance to 3.5.6.1, the vehicle shall be operated on dry level hard-surfaced road without loose material, and the results of 3 stopping tests shall be averaged to make this determination.

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4.5.1.28 Holding test. To determine conformance to 3.5.6.2, a simulated combat load shall be placed on the vehicle in a position that will not restrict the flow of engine air (intake or exhaust). The vehicle shall be driven up a 60 percent grade and the service brakes and then the parking brakes checked for conformance. The engine starting and fuel system tests on a 60 percent grade may be conducted in conjunction with this test.

4.5.1.29 Turning test. To determine conformance to 3.5.7, vehicle shall be operated and turned to right and to the left in a full pivot turn in neutral steer and checked for functional requirements.

4.5.1.30 Fording test. To determine conformance to 3.5.8.1, the vehicle shall be driven into water of specified depth without special equipment. During test period, water accumulation on the crew compartment hull floor, shall be not more than specified depth.

4.5.1.31 Turret and gun control system test.

4.5.1.31.1 Manual traverse effort. To determine conformance to 3.5.12.1.1, traverse mount through full range of 360° (6400 mils), in each direction, and observe operation. Effort shall not exceed specified requirements.

4.5.1.31.2 Traversing mechanism backlash. To determine conformance to 3.5.12.1.2, the traversing mechanism shall be checked as follows:

(a) If a backlash gage is available, check backlash as described in (1) through (4) below:

(1) Install scale component of backlash gage on bearing adapter.
(2) Install vernier component of backlash gage on output drive coupling and adjust knurled knob until gage reads zero.
(3) Twist coupling one way until gun begins to traverse; stop, and record reading on gage.
(4) Twist coupling in opposite direction until slack is taken up without moving the gun barrel; stop, and record the reading on the gage. A boresight or suitable telescope clamped to the gun and sighted on a distant object should be utilized to detect motion of gun barrel. Backlash shall not exceed 2 degrees and 5 minutes.

(b) If a backlash gage is not available, check backlash by first placing gun approximately horizontal, clamp a suitable collimator or telescope on breech casing and proceed as described in (1) through (5) below:

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(1) Attach a wire to azimuth oil gear shaft to serve as a pointer.
(2) Select a distant object, traverse gun either to right or left, and sight telescope so that the selected object appears at intersecting lines on the reticle. Approach object gradually and do not overpass.
(3) Without moving the gun, scribe a line on the 4 1/2 inch flange diameter of the bearing adapter to coincide with the wire pointer.
(4) Traverse gun in same original direction about one-half turn of the traversing hand crank.
(5) Reverse the direction of rotation until intersection of telescope reticle lines fall on original reference point but do not overpass. Position of wire pointer must coincide with scribed lines on bearing adapter within three thirty-seconds of an inch.

4.5.1.31.3 Power traverse. To determine conformance to 3.5.12.1.3, traverse gun mount slowly by moving the drive controller grip assemblies in a horizontal plane; with right grip assembly forward, the gun mount will traverse counterclockwise; with left grip assembly forward, the gun mount will traverse clockwise. Check for erratic movement or surge while traversing in either direction. Increase gun mount speed to full speed and check maximum speed with stop watch.

4.5.1.31.4 Elevation and traverse system stability. To determine conformance to 3.5.12.1.4, with the rate motor toggle switch on drive controller turned to the ON position, the gun and mount shall be checked for movement in traverse and elevation with stop watch, azimuth indicator and gunners quadrant. The stability of gun and mount shall be as specified.

4.5.1.31.5 Manual elevation effort. To determine conformance to 3.5.12.1.5, elevate and depress gun throughout full range of minus 3 degrees of plus 87 degrees and observe operation. Effort shall not exceed specified requirements.

4.5.1.31.6 Elevation mechanism backlash. To determine conformance to 3.5.12.1.6, the elevation mechanism shall be checked as follows:

(a) If a backlash gage is available, place gun in a horizontal position and check backlash as described in (1) through (4) below:

(1) Install scale component of backlash gage on bearing adapter.
(2) Install vernier component of the backlash gage on output drive coupling and adjust knurled knob until gage reads zero.
(3) Twist the coupling rubber seal one way until the gun barrel begins to move; stop, and record the reading on the gage.

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(4) Twist seal in opposite direction until slack is taken up and without moving the gun barrel; stop, and record reading on gage. A gunner's quadrant, boresight, or a suitable telescope clamped to the gun and sighted on a distant object should be utilized to detect motion of the barrel. Backlash shall be as specified.

(b) If a backlash gage is not available, check backlash by first placing gun approximately horizontal. Mount a telescope temporarily on breech casing and proceed as described in (1) through (5) below:

(1) Attach a wire to oil gear shaft to serve as a pointer.
(2) Select a distant object; elevate gun and sight telescope so that the selected object appears at intersecting lines on the reticle. Come up to the distant object gradually and do not overpass.
(3) Without moving the gun, scribe a line on the 5-3/8 inch diameter of the adapter housing to coincide with the wire pointer.
(4) Elevate gun approximately one-half turn of elevation hand crank.
(5) Depress gun slowly until the intersection of telescope reticle line falls on original reference point but do not overpass. Position of wire pointer must coincide with scribed line on housing within three-sixteenths of an inch.

4.5.1.31.7. Manual elevation limit stops. To determine conformance to 3.5.12.1.7, with the mount or carriage level the gun shall be operated to maximum depression and gun stop setting checked with gunners quadrant. The gun shall then be operated to maximum elevation and gun stop setting checked with gunners quadrant.

4.5.1.31.8 Power elevation. To determine conformance to 3.5.12.1.8, with mount or carriage level the guns shall be operated through the full range of travel and performance observed. The guns shall be brought smoothly to rest when slewed into either hydraulic elevation limit at a minimum speed of 25° per second and position of guns checked with gunners quadrant. Using stop watch operate guns in slew speed at specified positions and check for conformance to speed requirements.

CAUTION: Hydraulic stops shall be checked for proper setting before operating gun in full slew.

4.5.1.32 Fire control system test.

4.5.1.32.1 Synchronization. To determine conformance to 3.5.13.1, synchronization shall be checked as follows:

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(a) Place vehicle on as level terrain as possible and lock in azimuth.
(b) Prepare the gun mount for manual operation and make certain that the upper computer cover has been removed from the computing sight.
(c) Release gun and gun mount locks.
(d) Rotate the computer positioning handwheel of the computing sight until the plus 1,600 mil graduation on the azimuth scale of the computer housing is alined with the azimuth scale index.
(e) Set the speed knobs to the boresight mark ("B") at the 20 mph graduation on the knob scale and move the computer bail to 50 dive on the angle of flight scale.

(f) Place gunner's quadrant M1 or M1A1 on the gun mount carriage leveling pads parallel to the dual gun with the line of fire arrow pointing toward the gun muzzle.
(g) Adjust the gunners quadrant until the bubble in the level vial is centered. (This provides the basic setting for all the leveling procedures which follow.) Note the quadrant reading for future use.

NOTE: To prevent variations in reading during synchronization and/or boresighting, unnecessary movement of personnel on or over the vehicle, particularly the mount, is to be avoided.

(h) Place the gunners quadrant on the breech casing leveling pads. Manually level the dual gun to the quadrant setting obtained in (g) above.

NOTE: Be certain to check the micrometer reading on the gunners quadrant to insure that the original setting was not disturbed.

(i) Place gunners quadrant on computer body leveling pads and check level of computer body. This quadrant reading must be within 2 mils of quadrant reading obtained in (g) above. If reading is not within the prescribed limits, drag link rod must be adjusted.
(j) The gunners quadrant shall read level within plus or minus 2 mils when placed on the mount carriage leveling pads, breech easing leveling pads, and the computer body leveling pads.
(k) Manually elevate dual gun to 45° and check level of computer body. This reading must be level within plus or minus 2 mils.
(l) Fully elevate dual gun and check level of computer body. From 45° to full elevation, this reading must be level within plus or minus 4 mils.

4.5.1.32.2 Boresighting check. To determine conformance to 3.5.13.2, after dual gun has been synchronized, the accuracy of boresighting shall be checked as specified herein.

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a. With the upper computer cover removed, rotate the computer positioning handwheel to align the plus 1,600 mil graduation on the azimuth scale of the computer housing with the azimuth scale index.

b. Set the speed knobs of overhauled computing sights to the boresight mark ("BS") at 20 mph graduation on the knob scale, and set the computer bail at 50° full dive.

NOTE: When setting the computer bail on the computing sight, move the bail to 60° dive, then back to 50° dive. This will eliminate variations in readings which result from inherent backlash in the sight.

c. Place gunners quadrant M1 or M1A1 on the gun mount carriage leveling pads parallel to the dual gun and with line of fire arrow pointing toward the gun muzzle.
d. Adjust the gunner's quadrant until the bubble in the level vial is centered. (This provides the basic setting for all the leveling procedure which follows). Note the quadrant reading for future use.
e. Place the gunner's quadrant on the breech casing leveling pads. Manually level the dual gun to the quadrant setting obtained in "d" above.
f. Place the gunner's quadrant on the computer mechanism body leveling pads. If reading is not level within ± 1 mil, the vertical slide joint stem must be adjusted "g" and "h" below.
g. Loosen No. 5 x 1/2 roundhead screw which clamps computer cover to the base of computer body and drop cover to expose end of vertical slide and slide joint stem. Loosen 1/4 x 3/4 cap screw in vertical slide and screw slide joint stem in or out until computer mechanism body is level within ± 1 mil.
h. Tighten the 1/4 x 3/4 cap screw in vertical slide. Position computer cover on computer body and secure by tightening the No. 5 x 1/2 round head screw.

NOTE: Exercise care that dust and dirt do not blow into the computer assembly while the computer cover is removed. After leveling the computer mechanism body, and before the cover is replaced, the accessible portion of the vertical slide and slide joint stem should be carefully wiped with a clean, lint-free cloth and lubricated with a light film of petroleum base hydraulic fluid type 1, MIL-H-6083.

i. Set speed knob at zero MPH and set computer bail at 0°. These settings will tilt the computer mechanism body forward 9 mils ± 1 mil, less than level. This operation sets in the proper super-elevation.
j. Set the speed knobs to the RS mark and the computer bail to 50 dive.
k. With gunner's quadrant on the computer mechanism body leveling pads, check for level reading. If reading is not level within 1 mil, repeat "g" through "j" above.

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4.5.1.32.3 Boresighting procedure. Breech boresight M17-7692047 and muzzle boresight 5276694 shall be used for aligning the lines of sight of the sighting instruments with a line of sight through the axis of the gun bores upon a common aiming point no less than 1,000 yards distance as follows:

(a) Pull the hand operating level of the left gun back as far as possible and fasten in this position while another man performs the operations outlined (b) and (c) below.
(b) Open top cover of gun.
(c) Insert breech boresight M17 into chamber of left gun. Position boresight so the peep hole is up and press into place. Carefully move hand operating level to SAFE position.

CAUTION: Exercise extreme care when installing breech boresight to prevent injury by accidentally tripping the contractors.

(d) Install muzzle boresight in muzzle of left gun with the crossed wires horizontal and vertical and the red disk hanging in view.
(e) Set speed knobs to RS mark (20 MPH) and computer bail to 50° dive. Set azimuth scale to plus 1,600 mils.
(f) Sighting through the boresights, move the dual gun until it is carefully aimed at a well defined object not less than 1,000 yards distant.
(g) Install reflex sight M24C.
(h) Adjust reflex sight M24C so that its reticle pattern is accurately centered on the same distant aiming point (6) above.
(i) Remove muzzle boresight. To remove breech boresight, pull hand operating level out of SAFE position and back as far as it will move. Fasten level back, reach into breech, and withdraw breech boresight. Move hand operating level to firing position after breech has been completely cleared.
(j) Close top cover to lift gun.

4.5.2 Control tests. Control tests for maintaining control of those requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein.

4.5.2.1 Frequency. The first vehicle off the production line, or one of the first ten vehicles produced, and thereafter one per month or as specified, shall be checked in accordance with the following control tests.

4.5.2.2 Failure. If the vehicle selected fails to pass any of the control tests the Government inspector shall stop acceptance examination and testing on subsequent lots until such time as conditions causing the failure have been remedied.

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4.5.2.3 Contamination test. To determine conformance to 3.3.1.1, 3.3.1.2 and 3.5.8.2 the vehicle after the water fording test, shall be examined for contamination of lubricants of the transmission, engine final drive, and road wheels 2,3, 4 and one of their corresponding arms, on each side of the vehicle.

4.5.2.4 Stowed equipment test. To determine conformance to 3.3.2, all manufacturer and depot installed OVE shall be stowed (combat loaded) on the selected vehicle. All items shall fit with proper clearance in the spaces provided. The equipment shall be removed after the test. The OVE, used shall be of the latest production available to the contractor; and the contractor shall be permitted to unpackage and repackage (for domestic shipment only) equipment used for test.

4.5.2.5 Fire extinguisher test. To determine conformance to 3.3.9, the fire extinguisher system shall be functionally tested. One vehicle selected from the first ten produced shall have the fire extinguisher tested both sides, subsequent tests of one in each fifty testing alternately the internal and external controls.

4.5.2.6 Engine operating test (shallow water). To determine conformance to 3.5.8.1.1, with engine operating as specified, the first 5 vehicles shall be tested and observed for performance requirements. The flooding of the engine compartment shall be by an approved method.

4.5.2.7 Bilge pump water removal check. To determine conformance to 3.5.8.3, the vehicle shall have sufficient water in hull to operate bilge pump and check the rate of water removal with the specified discharge head.

4.5.3 Comparison tests (CT). The Government may select vehicles any time during the contract overhaul period and subject vehicles to tests listed on Table IV and Table V to reveal deficiencies of manufacture or design thay may reduce the effective operation of these items in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of vehicles shall be on a spot check basis. Comparison test vehicles shall be combat loaded with all OVE stowed in their applicable spaces to conform to 3.3.2. Vehicles selected shall not include any vehicles previously tested for conformance to 4.5.2 (Control Tests).

4.5.3.1 Test failure. Failure of any vehicle tested as specified in 4.5.3 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the contractor that corrective action has been taken to eliminate the conditions.

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Table V. 1600 mile test combat loaded
(Government Proving Grounds)

Course	Mileage & Speeds
Hard surface roads	200 miles at varying speeds up to maximum
Gravel and dirt roads	1000 miles at varying speeds up to maximum
Hilly cross country	200 miles at varying speeds up to maximum
Level cross country	200 miles at varying speeds up to maximum

4.5.3.2 Fuel system test. To determine conformance to 3.3.4, and 3.3.4.1, the vehicle shall be operated throughout all speed and gear ranges over cross country, hilly terrain, and on smooth, level hard surface roads, and vehicle observed for functional requirements. Prior to performing this test, the fuel tank shall be filled. The system shall be checked for leaks before and after a two mile road test. After road test the fuel tank shall be examined for foreign matter.

4.5.3.3 Exhaust system. To determine conformance to 3.3.5, the vehicle shall be "buttoned up" and the engine operated at 900 to 1000 rpm for a period of not less than 30 minutes. With all exhaust system in operation, the carbon monoxide (CO) content of the driver's compartment shall be analyzed.

4.5.3.4 Communications equipment test. To determine conformance to 3.3.13.2, communication system equipment shall be installed, subjected to tests and checked for voltage supply.

4.5.3.5 Sub-zero temperature tests. To determine conformance to 3.5.1, the vehicle, properly serviced and equipped, shall be subjected to applicable tests at the sub-zero temperatures specified.

4.5.3.6 High temperature operation tests. To determine conformance to 3.5.1, the vehicle, properly serviced and equipped, shall be subjected to the applicable tests at the high temperatures specified and observed for performance and heating conditions.-

4.5.3.7 Speed on grades test. To determine conformance to 3.5.3.2, the vehicle shall be operated on grades of 10 percent and 3 percent. While ascending these grades, it shall operate at the sustained speeds specified.

4.5.3.8 Trench crossing tests. To determine conformance to 3.5.9, the vehicle selected shall be driven at a speed not to exceed 5 mph without stalling over trenches as specified and observed for performance requirements. After the test, the vehicle shall be examined for damage.

4.5.3.9 Vertical obstacle test. To determine conformance to 3.5.10, the vehicle selected shall be driven at a speed not to exceed 5 mph without stalling over vertical obstacles as specified and observed for performance requirements. After the test, the vehicle shall be examined for damage.

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4.5.3.10 Climbing test. To determine conformance to 3.5.11, each vehicle selected shall be driven in forward and reverse gear up an approved 60 percent grade.

4.5.3.11 Radio suppression test. To determine conformance to 3.5.14, the vehicle shall be subjected to a radio interference suppression test.

4.5.3.12 Carbon dioxide distribution check. To determine conformance to 3.3.9, and 4.5.2.5 the fire extinguisher system shall be tested and the results examined for adequate distribution and control. With the engine idling at 700 to 800 rpm, and CO_2 concentration meters located at the approximate center of each perforated distribution tube, the internal controls shall be operated as specified, using force gage. The first shot shall actuate one bottle of CO_2. After a time delay of not more than 4 seconds the system shall begin discharging CO_2 into the engine compartment. After returning the handle to its original position, a second actuation shall discharge the remaining two bottles of CO_2. The CO_2 meters shall register a peak CO_2 concentration of at least 70% diminishing to zero within one minute after discharge. Removal and weighing of the CO_2 cylinders shall show a minumum of 11 pounds of CO_2 discharged.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Vehicle shall be processed for storage and shipment in accordance with MIL-G-45178, or as specified by procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The vehicle is intended for use by the Armed Forces as a combat vehicle.

6.2 Ordering data. Contractual documents will specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

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6.4 The following shall be used with the classification of defects (see Table III) for leaks:

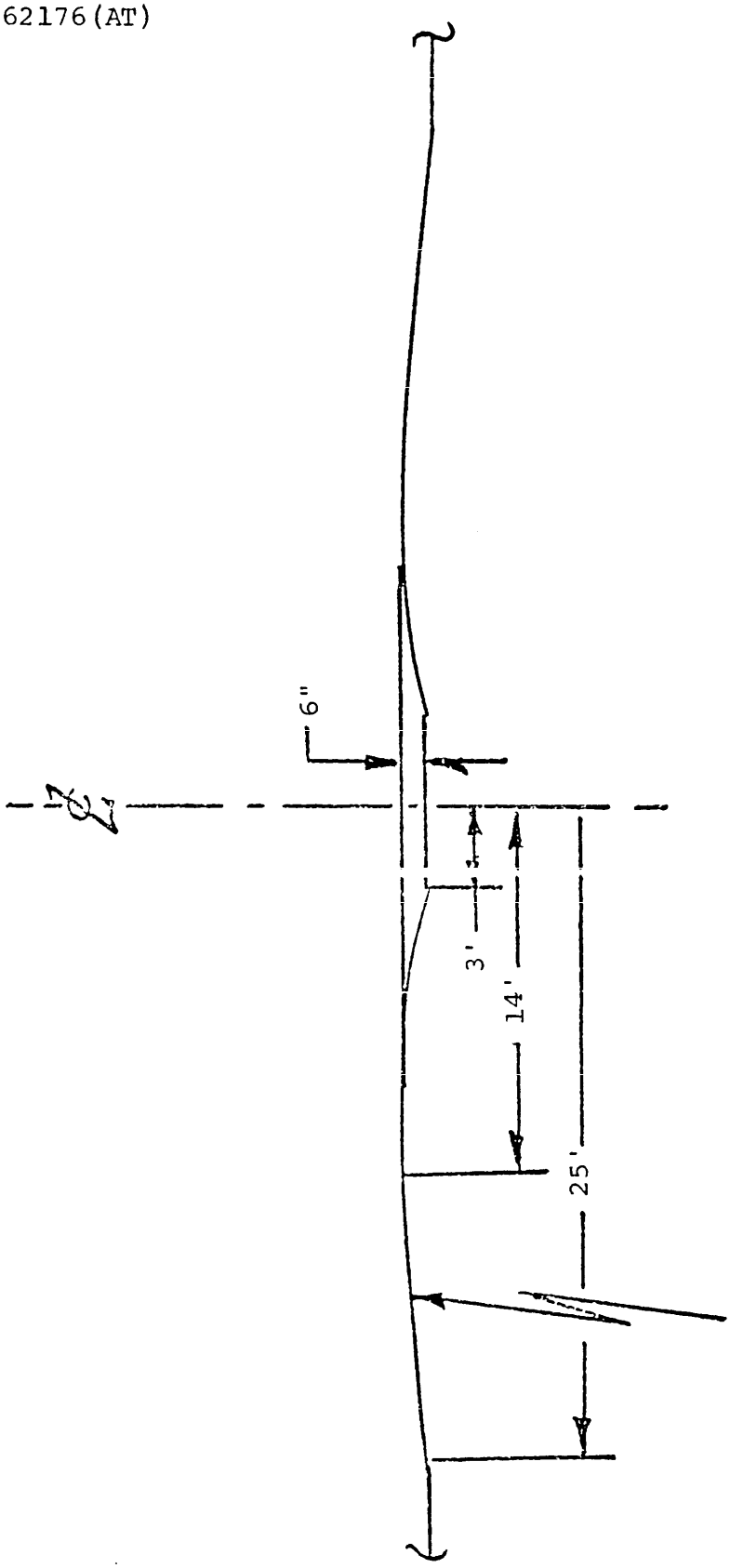
(a) Weep: Any evidence of fluid beyond the seal.
(b) Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.
(d) Drip: Any evidence of fluid beyond the seal where droplet forms and falls.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2350-A22

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CONTOUR APPROXIMATES
A 121 FT. RADIUS

FIGURE 1 - PROFILE OF TEST TRACK
(ARTIFICIAL BUMP)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-G-62176(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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